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Transparent by Design

Explainable AI in the Public Sector



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Why do we need explainability?

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The question

When software decides about people, can we explain the decision, and can they challenge it?

Who this is for

Students, researchers, programmers and product owners.

OPENING · ROADMAP

Five parts, one question

01

AI landscape

ANI, generative, agentic, AGI

02

Risks and bias

Where AI goes wrong, and why

03

Governance

Principles into testable controls

04

Explainable AI

How to build it, with code

05

Case Study and Takeaways

What we can learn from other country example?

The question: if an AI system decides about a citizen, can we explain why?

AI is a system, not just a model

WORKING DEFINITION

Machine-based systems that infer, generate, recommend or decide outputs for a given objective.

Govern by capability and risk, not by a perfect definition.

Model

The statistical or symbolic engine that produces outputs.

Data

Training, testing, live inputs, user and public data.

Infrastructure

Compute, cloud, chips, energy, APIs and tooling.

People + process

Who designs, deploys, supervises and challenges it.

Governance fails when it only governs the model.

ANI to AGI: the control question changes

ANI

Narrow AI

Optimizes one defined task: fraud flagging, translation, image triage.

Performance: is it accurate, fair, appealable?

GenAI

Generative AI

Creates new text, images, audio or code: chatbots, drafting, synthetic media.

Provenance: is it true, who made it?

Agentic

Agentic AI

Plans steps and uses tools or workflows: case routing, booking, procurement checks.

Permissions: what can it do, who approves?

AGI

The debate

General capability across tasks. Timelines are contested.

Preparedness: are we ready if it arrives?

Most AI in production today is still ANI. Govern capabilities, not predictions.

AI risk is not one risk

Bias and exclusion

UK A-level grades (2020): downgrades sparked reversal.

Control: bias tests, appeals

Opacity (our topic)

Australia Robodebt: income averaging, weak contestability.

Control: explanations, evidence trail

Security

EchoLeak in Microsoft 365 Copilot (2025): data exfiltration.

Control: prompt-injection tests

Power concentration

Cloud and model partnerships create lock-in risk.

Control: exit plans, interoperability

Public trust

Biden deepfake robocall (2024): voice impersonation.

Control: provenance, disclosure

Sustainability

Data centres: energy, water and e-waste impacts.

Control: energy and water reporting

Turn each risk into evidence: owner, threshold, control, metric, remedy.

Risk tiers decide what happens next

1	**Unacceptable**	**Prohibit or remove**	Manipulative systems, some biometric and social-scoring uses
2	**High**	**Allow with strong obligations**	Employment, education, health, essential public services
3	**Limited**	**Transparency or targeted controls**	Chatbots, deepfakes, AI-generated content labels
4	**Minimal**	**Light-touch governance**	Low-risk productivity and convenience tools

Essential public services sit in the high tier, so explainability is not optional there.

02 · BIAS

Where bias comes from, and what helps

Data bias

Training data reflects and amplifies historical inequities. Example: facial recognition misidentifying darker-skinned people.

WHAT HELPS

- Representative datasets
- Regular bias audits
- Synthetic data to fill gaps

Algorithmic bias

Design choices such as features, metrics and weak testing embed inequity, even when the data looks neutral.

WHAT HELPS

- Document features and metrics
- Test across demographic groups
- Diverse teams and affected users

Representation gap: only 26% of the computing workforce are women, and fewer than 20% of AI researchers and professors.

02 · OPACITY

The black box problem

If nobody can explain a decision, nobody can challenge it.

Dutch SyRI was shut down for being too opaque and violating citizens' rights.

Too complex to follow

Even the people who train the model may not understand it.

Behaviour that evolves

Systems keep learning and can behave unpredictably.

Hidden proxies

Facts are deduced from purchase history or geography.

Two ways out

1. Hold back deep learning in high-stakes use

2. Peer into the box: explainable AI

Principles need testable controls

Principle	Operational control	Evidence to request
Transparency	Notices, AI labels, user-facing explanation	Model cards, system logs, impact assessment
Human oversight	Approval gates, escalation, contestability	Reviewer sign-off logs, exception reports
Fairness	Bias testing on representative data	Results by group, remediation plans
Security	Access control, red teaming, incident response	Threat model, audit trail, red-team reports
Accountability	Named owner, audit rights, redress channel	RACI matrix, complaint records

A principle is not a control until someone can test it.

Four instruments that make oversight testable

1. Documentation

- Model and system cards
- Datasheets for datasets
- Public AI register
- Lineage and design logs

2. Explanations

- Plain-language notices
- Reason codes, counterfactuals
- SHAP / LIME attribution
- User tests of explanations

3. Accountability

- Named system owner
- RACI matrix: Responsible, Accountable, Consulted, Informed
- Human approval gates
- Audit rights, redress route

4. Monitoring

- Fairness dashboards
- Data and model drift
- Incident and complaint log
- Pause and rollback rules

Evidence pack: system card · dataset sheet · impact assessment · decision logs · fairness tests · audit trail

Public bodies are already doing this

Amsterdam, Helsinki	Public algorithm registers document AI used by city administrations (2020).
New York City	Public registry of AI models used in hiring, open to job seekers.
Netherlands, FRAIA	Mandatory fundamental-rights assessment when public bodies use algorithms.
Council of Europe	HUDERIA: map, assess, mitigate and monitor human-rights impact.
European Union	AI Act Article 50 transparency duties are now live.

What “explainable” actually means

Transparency
Openness about data, design and who owns the system.

Interpretability
A human can follow how the model works inside.

Explainability
We can give a reason for one specific outcome.

An explanation is always for someone:

Citizen
Plain reason and a way to appeal

Caseworker
Key factors to confirm or override

Auditor
Logs, tests and model versions

Developer
Debugging, drift and feature checks

Global explanation: how the model behaves overall. **Local** explanation: why this one case.

Two roads to an explainable model

Interpretable by design

Use a model a human can read.

- Logistic regression
- Decision trees and rules
- Explainable Boosting Machines

Trade-off: may give up some accuracy.

Explain a black box afterwards

Keep the complex model, add an explainer.

- SHAP: SHapley Additive exPlanations
- LIME: Local Interpretable Model-agnostic Explanations
- Surrogate models
- Counterfactuals

Trade-off: explanations approximate the model.

High stakes and tabular data? Start simple, and prove you need more.

Toolkits: AI Explainability 360 · SHAP · LIME · InterpretML · Fairlearn

Feature attribution with SHAP

```
import shap, xgboost

model = xgboost.XGBClassifier()
model.fit(X_train, y_train)

# one SHAP value per feature, per case
explainer = shap.TreeExplainer(model)
sv = explainer(X_test)

# local: why THIS applicant?
shap.plots.waterfall(sv[0])
# global: what drives the model?
shap.plots.beeswarm(sv)
```

Why this applicant? (illustrative)



Left: pushes toward **decline**. Right: pushes toward **approve**. Made-up numbers, not real data.

A postcode that moves the score is a question for the team, not just a bar on a chart.

Code: shap library, TreeExplainer on a tree model

From SHAP values to a notice a citizen can use

EXAMPLE DECISION NOTICE (ILLUSTRATIVE)

Your application was not approved.

Main reasons: two missed payments in the last 12 months; monthly income below the programme threshold.

What could change the outcome: with no missed payments over the next 6 months, the decision would likely change.

Your rights: ask a person to review this decision at [office / contact].

Reason codes

Top factors, ranked, in plain words.

Counterfactual

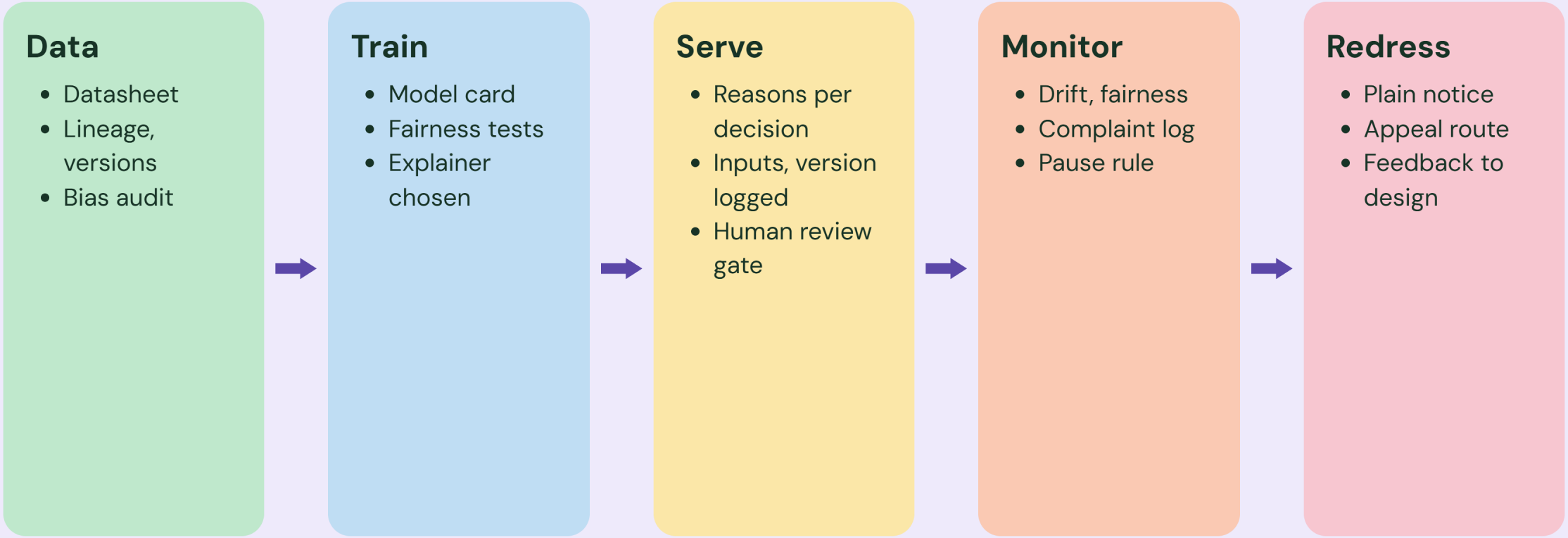
The smallest realistic change that flips the result. Never ask people to change protected traits.

Route to appeal

A named office and a human reviewer.

Notice wording is an example, not a legal template

Explainability is a lifecycle, not a plug-in



No stage moves forward without an owner, a control and a record.

Explanations can fail too

Unfaithful

A post-hoc explanation approximates the model and may not match its real logic.

Unstable

Small input changes can shift the attributions.

Proxy variables

Postcode or purchase history can stand in for protected traits. Seeing them is not a defence.

Explanation-washing

A tidy chart is not an audit. People may trust a wrong answer more when it comes with a reason.

Test fidelity and stability, test explanations with real users, keep a human in the loop.

What a transparent benefits service looks like

WHAT WENT WRONG IN REAL CASES

Robodebt (Australia): automated debt notices, weak contestability

SyRI (Netherlands): shut down for opacity and rights violations

A-levels (UK, 2020): statistical downgrades, public reversal

TRANSPARENT BY DESIGN MEANS

- Listed in a public AI register
- Impact assessment before launch
- Model card and dataset sheet
- Reason codes with every decision
- A caseworker who can override
- Appeal route and complaint log
- Monitoring with a pause threshold

TAKEAWAYS

Three things to remember

1 **Govern the whole system, not just the model.**

2 **A principle is real only when it is a testable control.**

3 **If it decides about people, it must be explainable and contestable.**

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Thank you

Questions? Let’s explain it together.

Scan to connect on LinkedIn

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